The Molecular Fock Projection: Diatomic Molecules as Prolate Spheroidal Lattices*

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Fock's 1935 stereographic projection maps the hydrogen atom onto the unit three-sphere S^3, where the Schrödinger equation becomes a pure Laplace–Beltrami eigenvalue problem with no potential term. The GeoVac program discretizes this S^3 as a graph Laplacian, achieving $< 0.1\%$ error for one-electron atoms with zero free parameters (Papers 0, 1, 7). For molecules, however, S^3 is not the natural geometry: the single-p_0 momentum scale cannot encode the R-dependent bonding physics (Paper 8–9, negative theorem).

We resolve this by identifying the *natural geometry principle*: every separable quantum system has a natural lattice—the lattice that discretizes the Laplace–Beltrami operator in the coordinate system where separation occurs. For one-center systems that geometry is S^3; for two-center systems it is the prolate spheroid. We discretize the separated prolate spheroidal equations for H_2^+ using a spectral angular solver (Legendre basis) and two radial solvers—a self-adjoint finite-difference (FD) scheme and a spectral Laguerre basis—with Brent root-finding in the separation parameter $c^2 = -R^2 E_{\mathrm{elec}}/2$ to enforce self-consistency. For σ states ($m = 0$), the spectral Laguerre matrix elements are computed algebraically from three-term recurrence relations, eliminating numerical quadrature entirely. For π and δ states ($m \neq 0$), an associated Laguerre basis absorbs the centrifugal $1/x$ singularity via partial-fraction decomposition and Stieltjes integral recurrence, reducing the non-algebraic content to a single transcendental seed value $e^a E_1(a)$. The method has *zero* free parameters: all inputs are nuclear charges, internuclear distance, and basis/grid resolution.

Results for H_2^+ with the spectral Laguerre radial solver ($n_{\mathrm{basis}} = 20$): minimum energy $E_{\min} = -0.6026$ Ha (0.0002% error vs. exact -0.6026 Ha), equilibrium bond length $R_{\mathrm{eq}} = 2.005$ bohr (0.38% error), and correct dissociation to H + p at $R \to \infty$. The spectral solver achieves a 250× dimension reduction, 270× speedup, and 5000× accuracy improvement over the FD solver, whose 1.01% error was a discretization artifact. This establishes the prolate spheroidal lattice as the molecular analog of Fock's atomic S^3 projection.

I. INTRODUCTION

The Geometric Vacuum program rests on a simple observation: the hydrogen atom's Schrödinger equation, when projected onto momentum space via Fock's 1935 conformal map [8], becomes the eigenvalue problem for the Laplace–Beltrami operator on the unit three-sphere S^3. The Coulomb potential $-Z/r$ is not an input force law but a coordinate distortion produced by the stereographic projection from S^3 to flat $\mathbb{R}^3$. This equivalence, established symbolically with 18 independent proofs [3], underlies the entire GeoVac computational framework: discretize S^3 as a graph, solve the graph Laplacian eigenvalue problem, and recover atomic energies to $< 0.1\%$ error with no fitted parameters [1, 2].

The extension to molecules has proven more subtle. Paper 8–9 established the *bond sphere* picture—a diatomic molecule as a single S^3 with two weighted poles—and derived the $SO(4)$ Wigner D-matrix elements that rotate atomic orbitals from one center to another [4]. However, that same paper proved a negative structural theorem: in the molecular Sturmian basis with a shared momentum scale p_0, the one-electron Hamiltonian and overlap obey the two-term identity $H_{ij} = -\frac{1}{2}p_0^2 S_{ij} + (1-\beta_j)V_{ij}$ and share a single orthogonal $SO(4)$ Wigner-D congruence (so H is block-, not scalar-, proportional to S), rendering the generalized eigenvalues independent of R. Binding in this framework requires the full R-dependent eigenvalue spectrum $\beta_k(R)$ from the continuous prolate spheroidal PDE—reintroducing the very continuous spatial geometry the graph was meant to replace.

For heteronuclear systems, the situation is worse: no shared p_0 exists that simultaneously represents both atomic length scales. The LCAO approach (atom-centered lattices with bridges) partially circumvents this, achieving a vertical binding energy $D_e^{\mathrm{CP}} = 0.110$ Ha for LiH at nmax= 3 ($\sim$ 20% error vs experiment) [5], but carries intrinsic BSSE and an R-independent kinetic energy, producing a monotonically attractive potential-energy surface with no equilibrium geometry.

This paper takes a fundamentally different approach. Rather than forcing molecules onto S^3, we ask: *what is the natural geometry for two-center systems?* The answer is immediate from classical mathematical physics: the two-center Coulomb problem separates in prolate spheroidal coordinates (ξ, η, φ), just as the one-center problem separates in spherical coordinates [9, 10]. Fock's projection exploits spherical separation via $SO(4)$ symmetry; the molecular analog exploits prolate spheroidal separation.

We formalize this as the **natural geometry principle**:

* Paper 11 in the Geometric Vacuum series

Every separable quantum system has a natural lattice—the lattice that discretizes the Laplace–Beltrami operator in the coordinate system where separation occurs.

For atoms, the natural lattice is the S^3 graph. For diatomics, it is the prolate spheroidal lattice. The GeoVac program is not about S^3 specifically—it is about identifying and discretizing the natural conformal geometry for each class of quantum systems.

The remainder of this paper implements this principle for H_2^+, the simplest diatomic. Section II reviews prolate spheroidal coordinates. Section III derives the separated equations. Sections IV and V describe the angular (spectral) and radial (finite-difference) solvers. Section VI introduces the spectral Laguerre radial solver that achieves 0.0002% accuracy. Section VII explains the self-consistency root-finding. Section VIII presents H_2^+ results: PES, spectroscopic constants, and grid convergence. Section IX discusses implications for the GeoVac program. Section X concludes.

II. PROLATE SPHEROIDAL COORDINATES

Place two nuclei A and B at the foci of a prolate spheroid, separated by distance R. The prolate spheroidal coordinates (ξ, η, φ) are defined by

$$\xi = \frac{r_A + r_B}{R}, \qquad \xi \in [1, \infty), \tag{1}$$

$$\eta = \frac{r_A - r_B}{R}, \qquad \eta \in [-1, +1], \tag{2}$$

$$\varphi = \arctan(y/x), \qquad \varphi \in [0, 2\pi), \tag{3}$$

where r_A and r_B are distances from the electron to nuclei A and B. Surfaces of constant ξ are confocal ellipsoids (with $\xi = 1$ the degenerate line segment joining the foci); surfaces of constant η are confocal hyperboloids ($\eta = \pm 1$ are the two half-axes beyond the nuclei).

The metric tensor has diagonal elements

$$g_{\xi\xi} = \frac{R^2}{4}\frac{\xi^2 - \eta^2}{\xi^2 - 1}, \quad g_{\eta\eta} = \frac{R^2}{4}\frac{\xi^2 - \eta^2}{1 - \eta^2}, \quad g_{\varphi\varphi} = \frac{R^2}{4}(\xi^2 - 1)(1 - \eta^2), \tag{4}$$

and the volume element is

$$dV = \frac{R^3}{8}(\xi^2 - \eta^2)\, d\xi\, d\eta\, d\varphi. \tag{5}$$

The nuclear distances take the simple forms

$$r_A = \frac{R}{2}(\xi + \eta), \qquad r_B = \frac{R}{2}(\xi - \eta), \tag{6}$$

so the nuclear attraction potential for charges Z_A and Z_B is

$$V = -\frac{Z_A}{r_A} - \frac{Z_B}{r_B} = -\frac{2}{R}\left[\frac{(Z_A + Z_B)\xi + (Z_B - Z_A)\eta}{\xi^2 - \eta^2}\right]. \tag{7}$$

The Laplacian in prolate spheroidal coordinates is

$$\nabla^2 = \frac{4}{R^2(\xi^2 - \eta^2)}\left[\frac{\partial}{\partial \xi}\left((\xi^2 - 1)\frac{\partial}{\partial \xi}\right) + \frac{\partial}{\partial \eta}\left((1 - \eta^2)\frac{\partial}{\partial \eta}\right) + \frac{\xi^2 - \eta^2}{(\xi^2 - 1)(1 - \eta^2)} \cdots \right. \tag{8}$$

III. SEPARATION OF THE TWO-CENTER COULOMB PROBLEM

The one-electron Schrödinger equation $(-\frac{1}{2}\nabla^2 + V)\Psi = E_{\mathrm{elec}}\Psi$ with the potential (7) can be written, after multiplying both sides by $-R^2(\xi^2 - \eta^2)/4$, as

$$\left[\hat{L}_\xi + \hat{L}_\eta + \frac{\xi^2 - \eta^2}{(\xi^2 - 1)(1 - \eta^2)}\frac{\partial^2}{\partial \varphi^2}\right]\Psi = 0, \tag{9}$$

where we define the ξ- and η-operators

$$\hat{L}_\xi = \frac{\partial}{\partial \xi}\left[(\xi^2 - 1)\frac{\partial}{\partial \xi}\right] + a\xi - c^2\xi^2, \tag{10}$$

$$\hat{L}_\eta = \frac{\partial}{\partial \eta}\left[(1 - \eta^2)\frac{\partial}{\partial \eta}\right] + b\eta + c^2\eta^2, \tag{11}$$

with the dimensionless parameters

$$a = R(Z_A + Z_B), \quad b = R(Z_B - Z_A), \quad c^2 = -\frac{R^2 E_{\mathrm{elec}}}{2}. \tag{12}$$

For bound states ($E_{\mathrm{elec}} < 0$), $c^2 > 0$.

The ansatz $\Psi(\xi, \eta, \varphi) = F(\xi)\, G(\eta)\, e^{im\varphi}$ separates Eq. (9) into two ordinary differential equations coupled through a separation constant A:

a. ξ-equation (radial):

$$\frac{d}{d\xi}\left[(\xi^2 - 1)\frac{dF}{d\xi}\right] + \left(A + a\xi - c^2\xi^2 - \frac{m^2}{\xi^2 - 1}\right)F = 0. \tag{13}$$

b. η-equation (angular):

$$\frac{d}{d\eta}\left[(1 - \eta^2)\frac{dG}{d\eta}\right] + \left(-A + c^2\eta^2 + b\eta - \frac{m^2}{1 - \eta^2}\right)G = 0. \tag{14}$$

The angular equation (14) is a generalization of the associated Legendre equation with additional $c^2\eta^2$ and $b\eta$ terms. For homonuclear diatomics ($Z_A = Z_B$), $b = 0$ and the η-equation reduces to the spheroidal wave equation. The radial equation (13) has an irregular singular point at $\xi = 1$ (the line segment between the nuclei) and an irregular singular point at $\xi \to \infty$.

The key feature of the separated system is that both equations (13) and (14) contain the same three unknowns: the energy E_{elec} (via c^2), the separation constant A, and the quantum numbers (m, n_ξ, n_η). For fixed m and state indices, the problem reduces to finding the value of c^2 for which both equations yield the same A.

IV. THE ANGULAR EQUATION: SPECTRAL SOLVER

The angular equation (14) is an eigenvalue problem for A at fixed c^2. We solve it using a spectral method with Legendre polynomial basis functions [11].

A. Legendre expansion

Expand $G(\eta) = \sum_{l=|m|}^{l_{\max}} a_l P_l^{|m|}(\eta)$ in associated Legendre polynomials. Substituting into Eq. (14) and using the recurrence relations for P_l^m produces a symmetric tridiagonal (pentadiagonal for $b \neq 0$) generalized eigenvalue problem

$$\mathbf{M}\,\mathbf{a} = A\,\mathbf{a}, \tag{15}$$

where the matrix $\mathbf{M}$ has elements determined by the three-term recurrence for $\eta^2 P_l^m$ and (when $b \neq 0$) ηP_l^m.

For homonuclear diatomics ($b = 0$), the matrix $\mathbf{M}$ is symmetric and tridiagonal, and the eigenvalues converge exponentially in $l_{\max}$. With $l_{\max} = 50$, the ground-state separation constant A is converged to machine precision for all c^2 values of interest.

B. Boundary conditions

The associated Legendre polynomials automatically satisfy the correct boundary conditions: regularity at $\eta = \pm 1$. No additional boundary conditions need to be imposed, because the factor $(1-\eta^2) \to 0$ at the endpoints provides a natural zero-flux condition for the self-adjoint operator.

For homonuclear diatomics, the solutions have definite parity under $\eta \to -\eta$: even parity (σ_g, π_u, ...) or odd parity (σ_u, π_g, ...). This parity is automatically captured by the Legendre expansion through the selection of even or odd l values.

V. THE RADIAL EQUATION: SELF-ADJOINT FINITE DIFFERENCES

The radial equation (13) is solved numerically on a uniform grid in ξ. The critical design choice is the discretization of the self-adjoint differential operator $d/d\xi\,[p(\xi)\,dF/d\xi]$ where $p(\xi) = \xi^2 - 1$.

A. Self-adjoint stencil

The standard second-order self-adjoint stencil [14] for $d/d\xi\,[p(\xi)\,dF/d\xi]$ on a uniform grid with spacing h evaluates p at the half-points:

$$\frac{1}{h^2}\left[p_{i+1/2}(F_{i+1} - F_i) - p_{i-1/2}(F_i - F_{i-1})\right], \tag{16}$$

where $p_{i\pm 1/2} = (\xi_i \pm h/2)^2 - 1$. This yields a symmetric tridiagonal matrix with diagonal elements

$$d_i = -\frac{p_{i+1/2} + p_{i-1/2}}{h^2} + A + a\xi_i - c^2\xi_i^2 - \frac{m^2}{\xi_i^2 - 1} \tag{17}$$

and off-diagonal elements

$$e_i = \frac{p_{i+1/2}}{h^2}. \tag{18}$$

B. Boundary conditions

Two boundary conditions are required:

a. At $\xi = 1$ (nuclear axis): For σ states ($m = 0$), the wavefunction is nonzero on the internuclear axis. Since $p(1) = 0$, the self-adjoint operator has a natural *Neumann* boundary condition at $\xi = 1$: the flux $p(\xi)\,dF/d\xi \to 0$ as $\xi \to 1^+$ regardless of the value of $dF/d\xi$. Computationally, this is implemented by setting $p_{1/2} = 0$ in the stencil (16), which modifies the first diagonal element to $d_1 = -p_{3/2}/h^2 + q_1$.

For π states ($m \neq 0$), the wavefunction vanishes on the axis by the $e^{im\varphi}$ factor, and a Dirichlet condition $F(1) = 0$ is appropriate.

b. At $\xi = \xi_{\max}$ (asymptotic region): A Dirichlet condition $F(\xi_{\max}) = 0$ is imposed, with $\xi_{\max}$ chosen large enough that the bound-state wavefunction has decayed to negligible amplitude. For H_2^+ near equilibrium, $\xi_{\max} = 25$ is sufficient; for large R, $\xi_{\max}$ is scaled with R.

C. Eigenvalue interpretation

With A fixed from the angular solver, the radial operator (13) is a Sturm–Liouville problem parameterized by c^2. At the *correct* c^2, the largest eigenvalue of the discretized operator equals zero—the bound state sits exactly at the top of the spectrum. At other values of c^2, the top eigenvalue is nonzero. This provides a scalar residual function $f(c^2) = \lambda_{\max}(c^2)$ for root-finding (Section VII).

The grid offset $\xi_{\min} = 1 + \delta$ (with $\delta = 5 \times 10^{-4}$) avoids the singular point $\xi = 1$ while keeping the first grid point close enough to capture the near-nuclear wavefunction.

VI. SPECTRAL LAGUERRE RADIAL SOLVER

The finite-difference radial solver of Sec. V converges as $O(1/N_\xi)$, requiring $N_\xi = 5000$ grid points for sub-percent accuracy. This section describes a spectral alternative that achieves 5000× better accuracy with 250× fewer degrees of freedom.

A. Laguerre basis

The radial equation (13) is defined on $\xi \in [1, \infty)$ with a regular singular point at $\xi = 1$ (where $p(\xi) = \xi^2 - 1 \to 0$) and exponential decay as $\xi \to \infty$. These features suggest a Laguerre basis:

$$\varphi_n(\xi) = e^{-\alpha(\xi-1)}\, L_n\big(2\alpha(\xi - 1)\big), \quad n = 0, 1, \ldots, N_b - 1, \tag{19}$$

where L_n is the Laguerre polynomial and the decay parameter $\alpha = \sqrt{c^2}$ is adapted to the wavefunction's asymptotic decay rate at each value of c^2.

B. Matrix elements

The kinetic operator $d/d\xi\,[(\xi^2 - 1)\, dF/d\xi]$ is treated via integration by parts:

$$\langle \varphi_m | \hat{T} | \varphi_n \rangle = -\int_1^\infty (\xi^2 - 1)\, \varphi_m'(\xi)\, \varphi_n'(\xi)\, d\xi, \tag{20}$$

where the boundary terms vanish: $(\xi^2 - 1) = 0$ at $\xi = 1$ and the exponential decay ensures convergence at $\xi \to \infty$. The potential matrix elements $\langle \varphi_m | q(\xi) | \varphi_n \rangle$ with $q(\xi) = A + a\xi - c^2\xi^2 - m^2/(\xi^2 - 1)$ and the overlap matrix $S_{mn} = \langle \varphi_m | \varphi_n \rangle$ are computed via Gauss–Laguerre quadrature, which is exact for the polynomial content of the integrands. For $m = 0$ (σ states), these matrix elements can alternatively be computed algebraically from Laguerre moment recurrences, eliminating quadrature entirely (Sec. VI E).

C. Generalized eigenvalue problem

The radial equation becomes the generalized eigenvalue problem

$$\mathbf{H}\, \mathbf{c} = \lambda\, \mathbf{S}\, \mathbf{c}, \tag{21}$$

where $\mathbf{H} = \mathbf{T} + \mathbf{Q}$ (kinetic plus potential) and $\mathbf{S}$ is the overlap matrix, both of dimension $N_b \times N_b$. The same Brent root-finding loop as in Sec. VII is used: at each trial c^2, the angular solver provides $A(c^2)$, the spectral radial solver provides $\lambda_{\max}(c^2, A)$, and the root $\lambda_{\max} = 0$ determines the energy. The spectral solver is activated via `radial_method='spectral'` with `n_basis=20`.

D. Spectral convergence

Table I compares the finite-difference and spectral radial solvers for H_2^+ at $R = 2.0$ bohr.

The spectral solver achieves a 250× dimension reduction, 270× speedup, and 5000× accuracy improvement over the FD solver. The FD solver's 1.01% error was a discretization artifact—the $O(1/N_\xi)$ convergence from the singular point at $\xi = 1$ and the first-order Neumann boundary condition (Appendix A) masked the intrinsic accuracy of the prolate spheroidal basis. The spectral solver, which treats the $\xi = 1$ boundary analytically via the vanishing of $(\xi^2 - 1)$ in the integration-by-parts identity (20), reveals that the prolate spheroidal separation itself is far more accurate than previously reported.

TABLE I. Comparison of finite-difference and spectral Laguerre radial solvers for H_2^+ at $R = 2.0$ bohr. Exact total energy: -0.6026 Ha.

Metric	FD ($N_\xi = 5000$)	Spectral ($N_b = 20$)
E_{total} error	1.01%	0.0002%
R_{eq} (bohr)	~ 2.01	2.005
R_{eq} error	0.65%	0.38%
Wall time (s)	4.3	0.016
Matrix dimension	5000	20

Spectral convergence is rapid: $N_b = 5$ already achieves sub-0.01% accuracy, and $N_b = 20$ reaches the 10^{-4}% level. The existing FD solver is preserved as the default for backward compatibility.

E. Algebraic matrix elements

1. σ states ($m = 0$)

For $m = 0$ (σ states), all matrix elements in the spectral Laguerre solver can be computed *algebraically* from three-term recurrence relations, eliminating Gauss–Laguerre quadrature entirely.

Define the *Laguerre moment matrices*

$$M_k[i,j] = \int_0^\infty x^k\, L_i(x)\, L_j(x)\, e^{-x}\, dx, \tag{22}$$

where L_n denotes the Laguerre polynomial of degree n. The three-term recurrence

$$x\, L_n(x) = -(n+1)\, L_{n+1}(x) + (2n+1)\, L_n(x) - n\, L_{n-1}(x) \tag{23}$$

gives M_k as finite band matrices built recursively from $M_0 = \mathbf{I}$ (orthonormality): M_1 is tridiagonal with $M_1[n,n] = 2n+1$, $M_1[n, n\pm 1] = -(n+1)$ or $-n$; M_2 is pentadiagonal, and so on. Higher moments are obtained by applying the recurrence (23) to reduce M_k to a sum of products involving M_{k-1}.

Under the substitution $x = 2\alpha(\xi - 1)$, each matrix element reduces to a finite linear combination of Laguerre moments:

- **Overlap:** $S_{mn} = \delta_{mn}/(2\alpha)$.
- **Potential:** The effective potential $q(\xi) = A + a\xi - c^2\xi^2$ (at $m = 0$, no singular term) becomes a quadratic in x, giving $\mathbf{V} = (2\alpha)^{-1}[q_0\, \mathbf{M}_0 + q_1\, \mathbf{M}_1 + q_2\, \mathbf{M}_2]$ where q_0, q_1, q_2 are determined by the coefficients A, a, c^2, and α.

- **Kinetic:** Writing $\varphi_n' = -2\alpha \sum_{j\leq n} L_j$ (from the Laguerre derivative identity) and using the integration-by-parts form (20), the kinetic matrix becomes $\mathbf{K} = -\frac{1}{2}\,\mathbf{F}\,\mathbf{M}_1\,\mathbf{F}^T - \frac{1}{8\alpha}\,\mathbf{F}\,\mathbf{M}_2\,\mathbf{F}^T$, where $F_{nj} = 2$ for $j < n$, $F_{nn} = 1$.

The resulting overlap and Hamiltonian matrices agree with Gauss–Laguerre quadrature to machine precision: maximum discrepancy 4.4×10^{-15} for the overlap and 4.1×10^{-12} for the Hamiltonian (the latter reflecting accumulated floating-point arithmetic in the moment products). PES energies agree to $< 2 \times 10^{-14}$ Ha across all bond lengths, confirming exact equivalence. The algebraic construction is approximately 1.6× faster than quadrature, as it avoids computation of Gauss–Laguerre roots and weights.

2. π, δ states ($m \neq 0$): associated Laguerre basis

For $m \neq 0$, the centrifugal term $m^2/(\xi^2 - 1)$ produces a $1/x$ singularity in the Laguerre variable $x = 2\alpha(\xi - 1)$. In the *ordinary* Laguerre basis, the required integral $\int_0^\infty L_i\, L_j\, e^{-x}/x\, dx$ diverges at $x = 0$—it has no finite Laguerre moment expansion.

The obstruction is removed by switching to the *associated Laguerre basis* $L_n^{(|m|)}(x)$ with weight $x^{|m|}e^{-x}$. The basis functions

$$\varphi_n(\xi) = (\xi - 1)^{|m|/2}\, e^{-\alpha(\xi-1)}\, L_n^{(|m|)}\big(2\alpha(\xi - 1)\big) \tag{24}$$

absorb the $(\xi-1)^{|m|/2}$ prefactor that enforces the Dirichlet condition at $\xi = 1$, converting the $1/x$ singularity into a *lowered moment* $M_{-1}^{(\alpha_L)}$ that converges for $\alpha_L = |m| \geq 1$.

The centrifugal matrix element decomposes via partial fractions:

$$\frac{m^2}{\xi^2 - 1} = \frac{m^2}{(x/\beta)(x/\beta + 2)} = \frac{m^2\beta}{2}\left(\frac{1}{x} - \frac{1}{x + 2\beta}\right), \tag{25}$$

where $\beta = 2\alpha$. The first term gives the lowered moment matrix $M_{-1}^{(\alpha_L)}$, computed algebraically from the summation identity (DLMF 18.9.13):

$$L_n^{(\alpha)}(x) = \sum_{k=0}^{n} L_k^{(\alpha-1)}(x) \quad\Longrightarrow\quad M_{-1}[i,j] = \sum_{p=0}^{\min(i,j)} h_p^{(\alpha_L-1)}, \tag{26}$$

where $h_p^{(\alpha)} = \Gamma(p+\alpha+1)/p!$ is the orthogonality constant.

The second term in (25) gives a *Stieltjes integral matrix*:

$$J_{ij}^{(\alpha_L)}(a) = \int_0^\infty \frac{x^{\alpha_L}\, e^{-x}\, L_i^{(\alpha_L)}(x)\, L_j^{(\alpha_L)}(x)}{x + a}\, dx, \tag{27}$$

with $a = 2\beta$. This is evaluated by a three-term recurrence seeded by the single transcendental value

$$S_0^{(0)}(a) = e^a\, E_1(a), \tag{28}$$

where E_1 is the exponential integral. Higher orders $S_0^{(k)}(a) = \Gamma(k) - a\, S_0^{(k-1)}(a)$ are rational in a for $k \geq 1$, and column entries follow from the associated Laguerre three-term recurrence applied to the denominator. Miller's backward recurrence is used to ensure numerical stability of the minimal solution.

The overlap, potential, and kinetic matrices in the associated basis are assembled from $M_0^{(\alpha_L)}$, $M_1^{(\alpha_L)}$, $M_2^{(\alpha_L)}$ (band matrices from the associated three-term recurrence), $M_{-1}^{(\alpha_L)}$, and $J^{(\alpha_L)}(2\beta)$. The kinetic matrix uses a derivative kernel with coefficients $G_0[n,n] = |m|/2 + n$, $G_0[n, n-1] = -(n+\alpha_L)$ and $G_1[n,n] = -1/2$, combined with weighted moment matrices $P_k = M_k + 2\beta\, M_{k-1}$.

Qualitative distinction from $m = 0$: The $m = 0$ algebraic construction is *fully rational*—every matrix element is a finite combination of integers and the Laguerre scale α. For $m \neq 0$, the single transcendental seed $e^a E_1(a)$ in (28) propagates through the Stieltjes matrix. All other matrix elements remain algebraic.

Table II shows the convergence behavior. The associated basis converges faster than the ordinary basis with quadrature for $|m| = 1$ (stable by $N = 10$ vs. non-monotonic at $N = 40$), and both bases agree to $\sim 5 \times 10^{-9}$ for $|m| = 2$.

TABLE II. H_2^+ $1\pi_u$ state ($m = 1$) at $R = 2.0$ bohr. Energies from the associated Laguerre (algebraic) and ordinary Laguerre (quadrature) spectral bases, compared to the FD reference (extrapolated). The associated basis converges by $N = 10$; ordinary Laguerre is non-monotonic and slower.

N	$E_{\rm assoc}$ (Ha)	$E_{\rm ord}$ (Ha)
5	0.071 228 3	0.072 583 7
10	0.071 228 2	0.071 156 5
15	0.071 228 2	0.070 794 9
20	0.071 228 2	0.070 659 0
25	0.071 228 2	0.070 770 1
30	0.071 228 2	0.070 844 7

VII. ROOT-FINDING AND SELF-CONSISTENCY

The complete algorithm proceeds as follows:

1. **Input:** Z_A, Z_B, R, grid parameters (N_ξ, $\xi_{\max}$), azimuthal quantum number m.

2. **Define residual** $f(c^2)$:

    (a) Compute $A(c^2)$ from the angular spectral solver (Section IV).

    (b) Compute $\lambda_{\max}(c^2, A)$ from the radial FD solver (Section V).

    (c) Return $f = \lambda_{\max}$.

3. **Bracket:** Find $c^2_{\min}$ and $c^2_{\max}$ such that $f(c^2_{\min}) > 0$ and $f(c^2_{\max}) < 0$. For homonuclear H_2^+, typical brackets are $c^2 \in [0.01, R^2(Z_A + Z_B)^2]$.

4. **Root-find:** Apply Brent's method [13] to solve $f(c^2) = 0$ to tolerance 10^{-12}.

5. **Extract energy:** $E_{\text{elec}} = -2c^2/R^2, \quad E_{\text{total}} = E_{\text{elec}} + Z_A Z_B/R$.

The algorithm is completely parameter-free: the only inputs are the physical quantities (Z_A, Z_B, R) and the numerical resolution (N_ξ, $\xi_{\max}$). No orbital exponents, basis set parameters, kinetic scaling factors, or bridge coupling constants appear anywhere.

Each evaluation of $f(c^2)$ requires one angular eigenvalue solve ($O(l_{\max})$ for tridiagonal) and one radial eigenvalue solve ($O(N_\xi)$ for tridiagonal). Brent's method typically converges in ~ 50 iterations, so the total cost per energy point is $O(50 \times N_\xi) \sim O(N_\xi)$. For $N_\xi = 5000$, a single point takes approximately 0.2 s on commodity hardware.

VIII. RESULTS: H_2^+ GROUND STATE

A. Grid convergence

Table III shows the convergence of the ground-state total energy for H_2^+ at $R = 2.0$ bohr as a function of radial grid size N_ξ (with $\xi_{\max} = 25$). The exact value [9] is $E_{\text{total}} = -0.6026$ Ha.

TABLE III. Grid convergence for H_2^+ at $R = 2.0$ bohr. The exact total energy is -0.6026 Ha.

N_ξ	c^2	E_{total} (Ha)	Error (%)
500	2.114	−0.5568	7.60
1000	2.156	−0.5779	4.11
2000	2.179	−0.5893	2.21
5000	2.193	−0.5965	1.01
10000	2.198	−0.5990	0.59

The convergence is approximately $O(1/N_\xi)$, consistent with second-order finite-difference discretization of the radial equation. Sub-percent accuracy is achieved at $N_\xi = 5000$ (0.2 s wall time). The spectral Laguerre radial solver (Sec. VI) eliminates this discretization bottleneck, achieving 0.0002% accuracy with $N_b = 20$.

B. Potential energy surface

Table IV gives the H_2^+ potential energy surface computed with $N_\xi = 8000$ over $R \in [1.0, 6.0]$ bohr. The curve exhibits a clear minimum near $R = 2.0$ bohr and dissociates correctly toward $E = -0.5$ Ha (the H atom ground state plus a bare proton) at large R.

TABLE IV. H_2^+ potential energy surface near the minimum ($N_\xi = 8000$, $\xi_{\max} = 25$).

R (bohr)	E_{total} (Ha)
1.50	−0.5783
1.60	−0.5869
1.70	−0.5926
1.80	−0.5961
1.90	−0.5979
2.00	−0.5984
2.10	−0.5979
2.20	−0.5965
2.50	−0.5894
3.00	−0.5730
4.00	−0.5412
5.00	−0.5190
6.00	−0.5058

C. Spectroscopic constants

Table V compares the computed spectroscopic constants with exact values. Equilibrium geometry and binding energy are obtained from a quadratic fit to the PES near the minimum.

TABLE V. Spectroscopic constants for H_2^+ ($N_\xi = 8000$). Exact values from Bates *et al.* [9].

Quantity	Computed	Exact	Error (%)
R_{eq} (bohr)	2.001	1.997	0.21
$E_{\min}$ (Ha)	−0.5984	−0.6026	0.70
D_e (Ha)	0.0984	0.1026	4.09
k (Ha/bohr^2)	0.1032	~ 0.10	~ 3

The equilibrium bond length is reproduced to 0.21% and the minimum energy to 0.70%. The dissociation energy $D_e = -0.5 - E_{\min}$ has a larger relative error (4.09%) because it is the small difference between two large numbers. The force constant k agrees to $\sim 3\%$.

D. Dissociation limit

At large R, the electronic energy approaches $E_{\text{elec}} \to -1/2$ (the hydrogen atom ground state) and the total energy approaches $E_{\text{total}} \to -1/2$ (since $V_{NN} = 1/R \to 0$). At $R = 10$ bohr with increased grid parameters ($N_\xi = 8000$, $\xi_{\max} = 80$), we obtain $E_{\text{total}} = -0.476$ Ha, approaching the exact limit of -0.500 Ha. The slow convergence at large R reflects the need for a large $\xi_{\max}$ to capture the diffuse wavefunction.

E. Heteronuclear extension: HeH^{2+}

The method extends naturally to heteronuclear one-electron systems. For HeH^{2+} ($Z_A = 2$, $Z_B = 1$) at

$R = 2.0$ bohr, we obtain $E_{\text{total}} = -1.512$ Ha (converged spectral solver, $n_{\text{basis}} \geq 20$; the FD-era -1.475 was under-resolved). The heteronuclear case introduces a nonzero $b = R(Z_B - Z_A)$ parameter in the angular equation, breaking the $\eta \to -\eta$ parity symmetry, but the algorithm handles this without modification.

a. How the η-label drifts with R. **[MEASURED]** The separation parameter A reduces to $l(l+1)$ in the united-atom limit, so the ground η-eigenvector's participation deficit $D = 1 - \max_l |v_l|^2$ in the Legendre basis measures how far the state has drifted from a pure united-atom l-label. The two cases turn on at different orders. For H_2^+ ($b = 0$) the $c^2\eta^2$ term couples $l \leftrightarrow l \pm 2$ and the deficit rises as $R^{4.0}$; for HeH^{2+} the $b\eta$ term couples $l \leftrightarrow l \pm 1$ and the deficit rises as $R^{2.1}$—first order in the coupling, the heteronuclear mixing being a factor R earlier. The large-R behavior differs too: the homonuclear deficit saturates near $\frac{1}{2}$ (gerade mixing of two Legendre components), whereas the heteronuclear deficit keeps rising (0.44 at $R = 2$, 0.71 at $R = 8$, 0.75 at $R = 12$) as $\langle\eta\rangle \to -1$: the label is lost to *localization* on the He focus, not to mixing. Both curves are smooth in R. This is the one-electron, continuous face of the abelian residue of Paper 58 [15]—m exact at every R, l replaced by a parameter that deforms rather than switches. Backing: `tests/test_paper11_eta_mixing_onset.py` (the two onset exponents, solver-free); the large-R values are the solver's, chronicled in CHANGELOG v5.10.2.

IX. DISCUSSION

A. The natural geometry principle

The results of this paper, combined with the atomic results of Papers 0, 1, and 7, establish a pattern:

- **One-center systems (atoms):** The natural geometry is the three-sphere S^3, accessed via Fock's stereographic projection from momentum space. The Schrödinger equation becomes the Laplace–Beltrami eigenvalue problem on S^3, with no potential term.
- **Two-center systems (diatomics):** The natural geometry is the prolate spheroid, where the two-center Coulomb problem separates into two coupled ODEs. The bond length R enters through the metric and the separation parameter c^2.

In both cases, the strategy is identical: identify the coordinate system where the Schrödinger equation separates, discretize the resulting Laplace–Beltrami operator, and solve the resulting eigenvalue problem. The physics—nuclear charges, distances, masses—enters only through the coordinate system and its metric. There are no basis set parameters, no fitted constants, and no orbital exponents to optimize.

B. Resolution of the bond sphere impasse

Paper 8–9 proved that the molecular Sturmian basis on a single S^3 with shared momentum scale p_0 cannot produce R-dependent binding: the one-electron (H, S) obey the two-term identity $H_{ij} = -\frac{1}{2}p_0^2 S_{ij} + (1-\beta_j)V_{ij}$ and share a single $SO(4)$ Wigner-D congruence, making the eigenvalues geometry-independent. The prolate spheroidal lattice resolves this impasse by abandoning the single-center S^3 geometry entirely. Rather than forcing two centers onto one sphere, each center retains its identity within the natural two-center geometry.

The bond length R does not enter as an external parameter grafted onto an R-independent topology. Instead, R is intrinsic to the prolate spheroidal metric (4): it sets the interfocal distance, which determines the coordinate system itself. This is analogous to how p_0 is intrinsic to the Fock projection for atoms—it is not a free parameter but is determined self-consistently by the energy eigenvalue.

C. Comparison with LCAO approach

The LCAO approach [5] constructs a molecular basis by combining atom-centered lattices with bridge elements. While it gives a vertical binding energy ($D_e^{\text{CP}} = 0.110$ Ha for LiH, $\sim 20\%$ error, no equilibrium geometry), it has inherent limitations:

1. **BSSE:** Combining two lattices creates basis set superposition error (-0.115 Ha for LiH at nmax= 3), requiring counterpoise correction.
2. **R-independent kinetic energy:** The graph Laplacian eigenvalues on each atomic lattice do not depend on R, shifting R_{eq} to ~ 2.5 bohr (expt. 3.015).
3. **Bridge parameters:** The inter-atomic coupling requires STO overlap integrals and Mulliken-type approximations.

The prolate spheroidal lattice has none of these limitations: no BSSE (single coordinate system), R-dependent metric (correct R_{eq}), and no fitted parameters. For one-electron systems, it is clearly superior.

D. Limitations and extensions

The most important limitations and implemented extensions are:

a. Two-electron systems (H_2). We have implemented a prolate spheroidal CI using single-particle orbitals (σ_g, σ_u) from the separated H_2^+ solver and a 2×2 CI in the $^1\Sigma_g^+$ sector. The V_{ee} matrix elements are computed via azimuthal averaging of $1/r_{12}$ using the complete elliptic integral $K(k)$ on a two-dimensional (ξ, η) quadrature grid.

With frozen H_2^+ orbitals ($Z_{\rm eff} = 1$), the CI yields $D_e = 0.077$ Ha (44% of exact 0.175 Ha) at $R_{\rm eq} = 1.77$ bohr. Optimizing the orbital exponent via the Eckart variational method ($Z^*_{\rm eff} \approx 0.765$ at $R = 1.4$ bohr) reduces the Coulomb self-repulsion J_{gg} from 0.80 to 0.68 Ha and improves the result to $D_e = 0.101$ Ha (58% of exact) with $R_{\rm eq}$ matching the exact value of 1.401 bohr within fitting precision.

An important structural result is the *pi orbital selection rule*: $\langle \sigma_g \sigma_g | \pi_+ \pi_- \rangle = 0$ by azimuthal symmetry, so π orbitals cannot improve the ${}^1\Sigma_g^+$ ground state via CI mixing. The 4σ CI (6×6 matrix) confirms this, adding only 0.1 mHa over the 2×2 result.

The remaining 42% D_e error arises from the minimal two-orbital CI basis and missing dynamic correlation. The V_{ee} integrals converge as $O(1/N_{\rm grid}^2)$ in grid size.

b. Heteronuclear systems (HeH$^+$). For heteronuclear diatomics, we have implemented independent per-atom orbital exponent optimization ($Z_{{\rm eff},A}$, $Z_{{\rm eff},B}$), which recovers binding in HeH^+—a system that is unbound with frozen HeH^{2+} orbitals due to excessive Coulomb repulsion ($J_{11} \sim 1.3$ Ha).

c. Excited states. The current solver targets the ground σ_g state. Excited states (σ_u, π_g, π_u) are accessible by changing the azimuthal quantum number m and selecting different eigenvalues of the angular and radial operators.

d. Higher-order discretization. The spectral Laguerre radial solver (Sec. VI) has replaced the $O(1/N_\xi)$ FD scheme as the high-accuracy option, achieving 0.0002% error with $N_b = 20$ basis functions—a 5000× accuracy improvement over the FD solver at 250× smaller matrix dimension. For σ states, all matrix elements are now algebraic (Sec. VI E), with no remaining numerical quadrature in the radial solver.

X. CONCLUSION

We have demonstrated that the natural geometry for diatomic molecules is the prolate spheroid, just as the natural geometry for atoms is the three-sphere S^3. By discretizing the separated prolate spheroidal equations with a spectral angular solver and a spectral Laguerre radial solver, we obtain the H_2^+ ground-state energy to 0.0002% accuracy and the equilibrium bond length to 0.38% accuracy with zero free parameters. The spectral solver achieves a 250× dimension reduction and 270× speedup over the finite-difference radial solver, whose 1.01% error was a discretization artifact. For σ states, the spectral matrix elements are computed algebraically from Laguerre moment recurrences, eliminating numerical quadrature from the radial solver entirely.

This result establishes the *natural geometry principle* as the organizing idea of the GeoVac program:

> *Every separable quantum system has a natural lattice—the lattice that discretizes the Laplace–Beltrami operator in the coordinate system where separation occurs.*

The atomic S^3 lattice (Papers 0, 1, 7) and the molecular prolate spheroidal lattice (this paper) are the first two instances of this principle. The two-electron extension to H_2 via prolate spheroidal CI achieves 58% of the exact binding energy with a single variationally optimized orbital exponent and no empirical parameters. Future work will pursue larger CI expansions, polyatomic systems with other natural coordinate systems, and integration of the spectral radial solver into the full PES scan and composed-geometry pipelines.

The prolate spheroidal lattice also resolves the impasse identified in Paper 8–9: the single-p_0 Sturmian basis on S^3 cannot produce R-dependent binding, but the prolate spheroidal coordinate system encodes R directly in its metric, producing the correct equilibrium geometry without any fitted parameters.

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Appendix A: Grid Convergence Analysis

The second-order self-adjoint finite-difference stencil gives truncation error $O(h^2) = O((\xi_{\max}/N_\xi)^2)$. Since the top eigenvalue of the discretized radial operator must equal zero, the energy error is controlled by the eigenvalue error, which scales as $O(h^2)$ for the largest eigenvalue.

In practice, the observed convergence rate is closer to $O(1/N_\xi)$ (Table III) rather than $O(1/N_\xi^2)$. This sub-optimal convergence is attributed to:

1. The grid offset $\delta = 5 \times 10^{-4}$ from the singular point $\xi = 1$, which introduces an $O(\delta)$ boundary error.
2. The Neumann boundary condition implementation, which is first-order at the boundary.
3. The nonuniform behavior of $p(\xi) = \xi^2 - 1$ near $\xi = 1$, where $p \to 0$.

The spectral Laguerre solver of Sec. VI eliminates all three issues: the integration-by-parts identity (20) handles the $\xi = 1$ singularity analytically, and the exponential basis functions match the wavefunction's asymptotic decay, achieving 0.0002% accuracy with $N_b = 20$.

Appendix B: Comparison with Failed Approaches

Before arriving at the separated solver, two other discretization strategies were attempted and found inadequate.

1. 1D axial lattice

The simplest possible molecular lattice places vertices along the internuclear axis at $z \in \{0, R/4, R/2, 3R/4, R\}$ with nearest- neighbor edges. This fails catastrophically because:

1. The 1D Coulomb potential $-Z/|z|$ is too singular without the r^2 Jacobian factor from 3D integration.
2. The grid spacing $h = R/4$ artificially couples the kinetic energy scale to R.
3. The 1D model has no physical correspondence to the 3D molecular problem.

A cylinder-averaged variant (averaging the Coulomb potential over a cylinder of radius ρ_0) produces qualitatively correct PES shapes but requires ρ_0 as a free parameter, achieving only $\sim 30\%$ energy error at best.

2. 2D tensor-product grid

A direct 2D discretization on an $N_\xi \times N_\eta$ tensor product grid in (ξ, η) space solves the *unseparated* problem $H\Psi = EM\Psi$ where $M = \mathrm{diag}(\xi^2 - \eta^2)$ is the volume weight. While correct in principle, this approach suffers from:

1. Slow convergence: 28% error at $50 \times 25 = 1250$ unknowns, versus 1.01% error at $N_\xi = 5000$ for the separated solver.
2. Boundary condition difficulties: Dirichlet BCs at $\eta = \pm 1$ force $\Psi = 0$ at the nuclei, which is wrong for σ_g states. The correct natural zero-flux BCs are nontrivial to implement on a 2D grid.
3. Computational cost: the 2D generalized eigenvalue problem is $O(N^3)$ in the total number of grid points, versus $O(N_\xi)$ for the separated tridiagonal solver.

The separated approach is $\sim 50\times$ more efficient because it exploits the exact separability of the two-center Coulomb problem.

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